

2019 Acer Aspire 5 Hardware Teardown, System Rebuild, and Active Directory Homelab Project

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Date: January 1st, 2026

1. Executive Summary

This report documents the successful teardown, reassembly, system rebuild, and virtualization-based Active Directory deployment performed on a 2019 Acer Aspire 5 laptop. The purpose of this project was to build and demonstrate hands-on technical skills across a range of information technology concepts, including computer hardware, operating system deployment, virtualization, and system administration. The project involved safely disassembling and reassembling a 2019 Acer Aspire 5 laptop to identify and document internal components, performing a clean installation of Windows 11 using an official installation key, and deploying a functional Active Directory domain environment using VMware Workstation and Windows Server 2022.

2. Project Objectives

The objectives of this project were to:

- **Demonstrate hardware teardown and reassembly procedures**
 - **Identify and document internal laptop components and explain their functions**
 - **Perform a clean Windows 11 installation using proper firmware and boot configuration**
 - **Configure hardware-assisted virtualization**
 - **Deploy a working Active Directory domain environment using Windows Server 2022**
 - **Implement organizational units, users, security groups, and Group Policy Objects**
 - **Validate domain functionality using command-line tools**
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3. Hardware Teardown and Component Analysis

3.1 Laptop Disassembly

The project began with a complete teardown of the Acer Aspire 5 laptop. The system was powered off, disconnected from external power, and opened in accordance with standard hardware safety practices. The internal battery was disconnected before handling other components to minimize electrical risk.

The following internal components were identified, removed, and documented with photographs:

- **128 GB SK Hynix Solid State Drive**
- **4GB SK Hynix RAM Module**
- **Qualcomm Atheros Wifi card**
- **Cooling Fan**
- **Acer AC14B8K Lithium Ion Battery Pack**



128 GB Solid State Drive designed for high performance and portability



Acer AC14B8K Lithium Ion Batter Pack. Primary Source of power



Cooling fan used to prevent overheating and mitigate poor performance



Laptop Interior post-teardown. The motherboard was kept in place as a safety precaution.



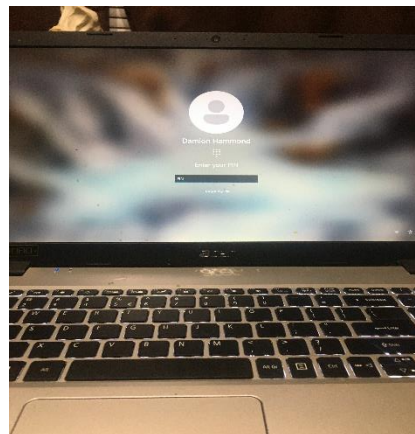
Due to the absence of model-specific laptop repair tools at the time of execution, alternative application repair tools were utilized to complete the hardware teardown safely. These included a plastic spudger, precision screwdrivers, and non-conductive tongs.

3.2 Component Function Descriptions

- **Solid-State Drive (SSD):** Primary storage device responsible for hosting the operating system, applications, and user data.
 - **RAM:** Temporary memory used by the system to support active processes and multitasking.
 - **Wi-Fi Card:** Provides wireless network connectivity for the laptop.
 - **Cooling System:** Regulates system temperature and prevents overheating during sustained workloads.
 - **Motherboard:** The main circuit board that interconnects all system components and facilitates communication between them.
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4. Laptop Reassembly

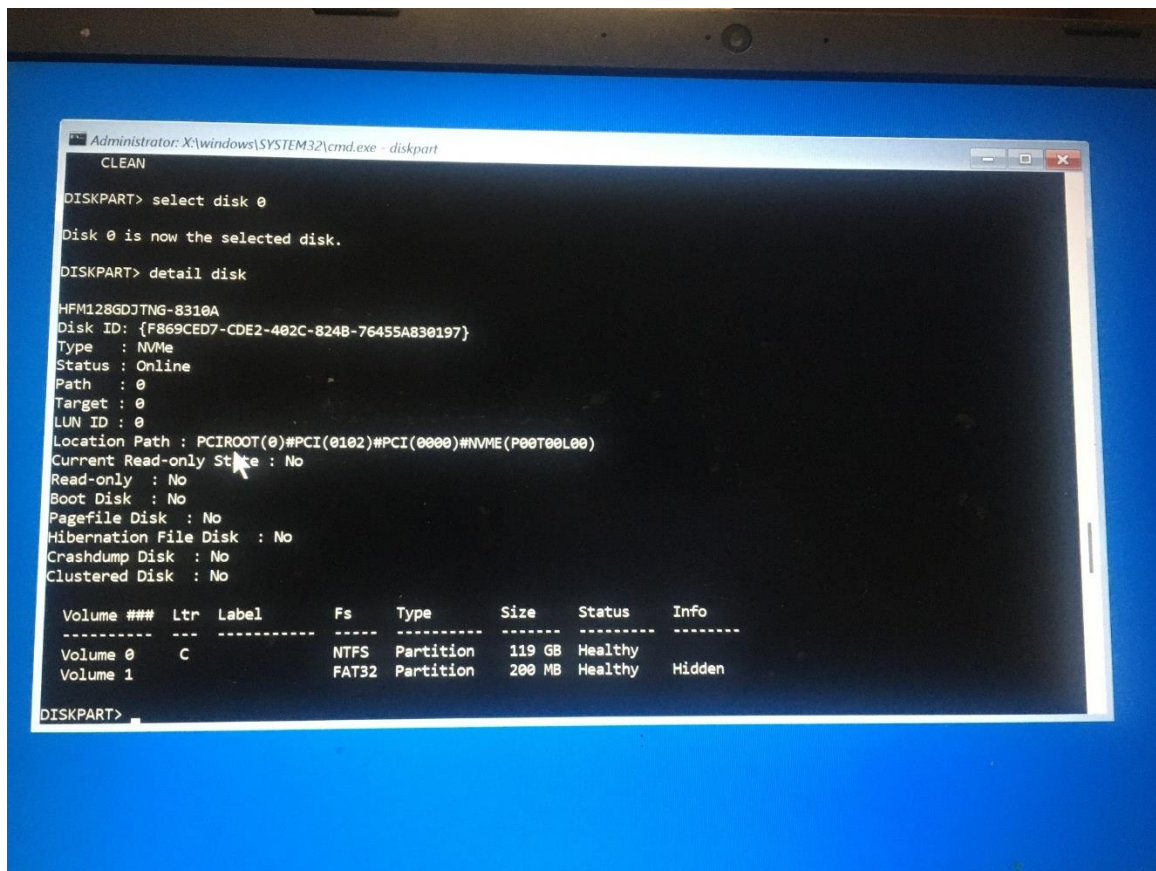
Following component documentation, the laptop was reassembled by reseating all internal components and reconnecting all ribbon cables and connectors. The cooling system, storage, memory, and battery were all secured in their original positions. After reassembly, the laptop was successfully powered on, confirming that all components were properly reinstalled and functioning as expected.



5. Operating System Reinstallation and Configuration

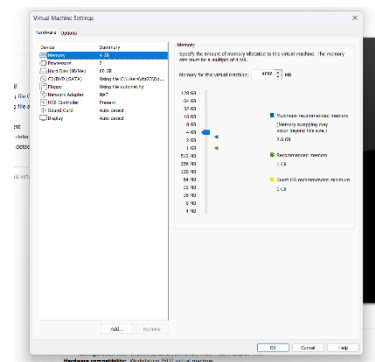
A clean installation of Windows 11 was performed using official Microsoft installation media and a valid installation key. The system firmware was configured to use UEFI boot mode, and existing partitions were removed to ensure a fresh operating system deployment. The installation process included:

- Disk preparation and partition creation
- Windows 11 installation and initial configuration
- Driver installation and system updates
- Verification of system stability and performance



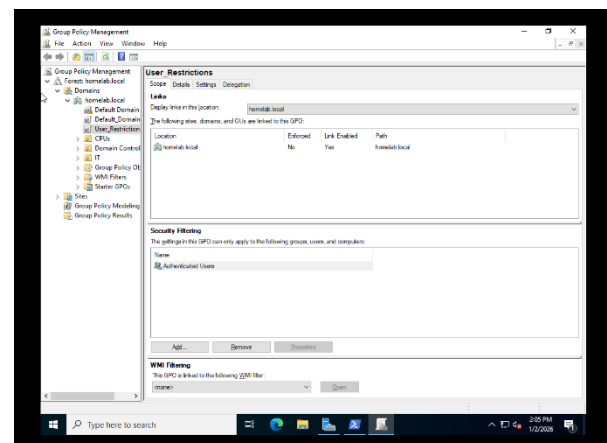
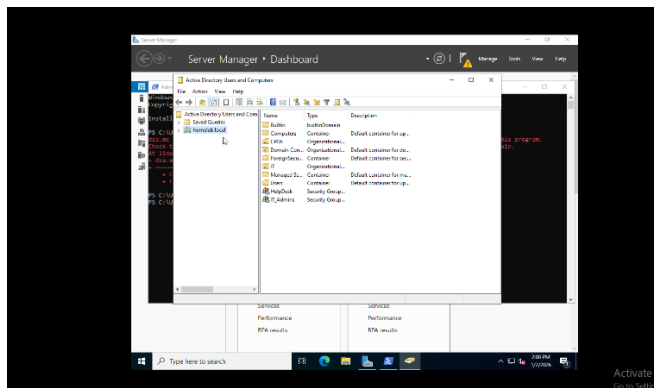
6. Virtualization Environment Setup

Hardware-assisted virtualization was enabled in the system firmware using AMD Secure Virtual Machine (SVM) support. VMware Workstation Pro was installed on the Windows 11 host system to serve as the virtualization platform. From there, the primary virtual machine was configured with appropriate CPU, memory, storage, and networking resources to simulate an enterprise environment while maintaining host system performance.



7. Active Directory Deployment:

A Windows Server 2022 virtual machine was deployed and configured as a domain controller. The server was assigned a static IP address and promoted to a domain controller within a newly created forest. The domain was named **homelab.local**, two services were installed (Active Directory Domain Services and Domain Name Services system), organizational units (OUs) were created to support structured administration (IT, Users, and Computers), and Group Policy Objects (GPOs) were implemented to enforce domain-level security and user restrictions. Policies included password length and complexity enforcement, account security and authentication, and user configuration restrictions (e.g., Control Panel access limitations).

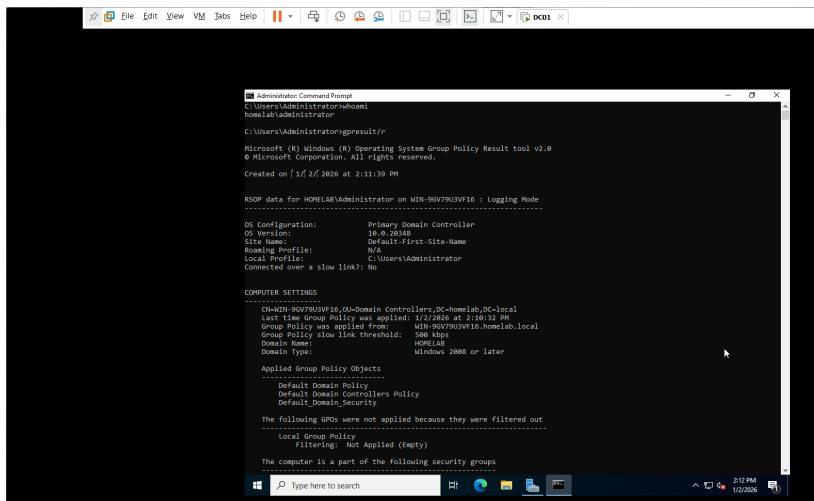


8. Validation and Testing:

The Active Directory environment was validated using a domain-joined client virtual machine. Successful testing included:

- **Domain join confirmation**
- **User authentication testing**
- **Group Policy enforcement verification**
- **Directory and DNS functionality checks**

Command-line tools such as **whoami**, **gpupdate /force**, and **gpresult /r** were used to confirm correct domain operation.



```
C:\Users\Administrator>whoami
HOMELAB\Administrator

C:\Users\Administrator>gpresult /r

Microsoft (R) Windows (R) Operating System Group Policy Result tool v2.0
© Microsoft Corporation. All rights reserved.

Created on 1/2/2026 at 2:11:39 PM

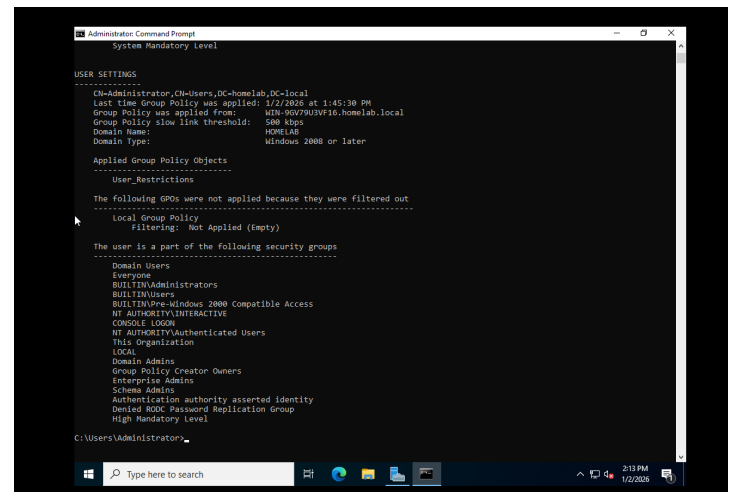
RDP data for HOMELAB\Administrator on MIN-90V79U3VF16 : Logging Mode
-----
OS Configuration:      Primary Domain Controller
OS Version:            10.0.22H2
Site Name:              Default-First-Site-Name
Roaming Profile:        N/A
Local Profile:          C:\Users\Administrator
Connected over a slow link? No

COMPUTER SETTINGS
-----
OS: MIN-90V79U3VF16-01-Domain Controllers,DC-HOMELAB,DC-local
Last time Group Policy was applied: 1/2/2026 at 2:10:32 PM
Group Policy was applied from: MIN-90V79U3VF16-HOMELAB.local
Group Policy slow link threshold: 500 kbps
Domain Name:           HOMELAB
Domain Type:            Windows 2008 or later

Applied Group Policy Objects
-----
Default Domain Policy
Default Domain Controllers Policy
Default Domain Security

The following GPOs were not applied because they were filtered out
-----
Local Group Policy
Filtering: Not Applied (Empty)

The computer is a part of the following security groups
-----
```



```
System Mandatory Level

USER SETTINGS
-----
CH-Administrator,CH-Users,DC-HOMELAB,DC-local
Last time Group Policy was applied: 1/2/2026 at 1:45:30 PM
Group Policy was applied from: MIN-90V79U3VF16-HOMELAB.local
Group Policy slow link threshold: 500 kbps
Domain Name:           HOMELAB
Domain Type:            Windows 2008 or later

Applied Group Policy Objects
-----
User_Restrictions

The following GPOs were not applied because they were filtered out
-----
Local Group Policy
Filtering: Not Applied (Empty)

The user is a part of the following security groups
-----
Domain Users
Everyone
BUILTIN\Administrators
BUILTIN\Users
BUILTIN\Win-2000-Compat-Access
NT AUTHORITY\INTERACTIVE
CONSOLE LOGON
NT AUTHORITY\Authenticated Users
This Organization
LOCAL
Domain Admins
Group Policy Creator Owners
Enterprise Admins
Schema Admins
Authentication authority asserted identity
Denied RODC Password Replication Group
High Mandatory Level

C:\Users\Administrator>
```

9. Challenges and Troubleshooting:

Throughout the execution of this project, multiple technical challenges were encountered and required systematic troubleshooting, research, and configuration adjustments to resolve. Accessing the system firmware (BIOS/UEFI) presented an initial challenge because standard function key methods such as F2/F12 did not consistently allow entry into the BIOS due to the fast boot behavior and OEM firmware restrictions. This issue was resolved by entering the firmware via Windows Advanced Startup and disabling the fast boot feature, which enabled reliable access for configuration changes. Additionally, secure boot settings and UEFI boot mode required adjustment to support a clean operating system installation and virtualization.

On the Acer Aspire 5 platform, virtualization options were not immediately visible within the firmware. The system required the creation of a temporary supervisor password to unlock advanced BIOS settings. Once unlocked, AMD-specific virtualization features such as AMD-SVM and AMD-IOMMU were identified and enabled. This highlighted the importance of understanding OEM-specific firmware behavior when configuring hardware-assisted virtualization. Finally, during the Windows Server 2022 installation in VMware Workstation Pro, a licensing error indicated that the Microsoft Software License Terms could not be found. This issue was traced to insufficient memory allocation within the virtual machine. The problem was resolved by increasing the VM's RAM allocation to meet minimum installation requirements and verifying that the correct ISO was mounted and connected.

Across all challenges, the standard troubleshooting procedure was applied to ensure success: Identify the problem, gather information, identify potential causes, test and isolate, and implement solutions.

10. Conclusion:

This project successfully demonstrated end-to-end IT systems administration skills, from handling physical hardware to deploying an Active Domain Directory environment. The completed Acer Aspire 5 homelab environment closely mirrors real-world IT tasks and serves as a strong foundation for future infrastructure and cloud-based projects. This project reinforced the interdependencies among hardware, firmware, operating systems, virtualization platforms, and directory services. Minor configuration issues at the firmware or network level can significantly impact higher-level services such as Active Directory. This project also emphasized the importance of clear documentation, validation, and structured administration when managing system infrastructure.